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A MULTI-ECHELON INVENTORY MODEL FOR REPAIRABLE ITEMS WITH EMERGENCY LATERAL TRANSSHIPMENTS*

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In multi-echelon inventory systems that cover extensive geographical regions, emergency lateral transshipment between bases is commonly practiced to provide improved service support for products. In these systems, since neighboring bases are at shorter distances than to the depot, the base may seek stock from them to satisfy its demands when it is out of stock. In this paper we develop a continuous-review multi-echelon model for repairable items when emergency lateral transshipments between identical bases are allowed. Approximations are derived and tested for the expected level of backorders and the quantity of emergency lateral transshipments in cases where fairly high service levels are required. These approximations are used to determine the optimal stocking levels in such a system. A procedure to find these stocking levels is given. Numerical examples illustrate the application of the procedure and the tradeoffs involved in emergency lateral transshipment.
(MULTI-ECHELON MODELS; APPROXIMATION IN INVENTORY MODELS; ORDERING POLICIES)

1. Introduction

Multi-echelon inventory systems are usually used to provide service support for products whose customers are distributed over an extensive geographical region. For repairable items, stocking of the product enables the system to provide operative items when there are demands for them (generated by the failure of the items). These systems are usually characterized by a lower echelon, consisting of bases that serve as a first level of product support, and a higher echelon with a depot that serves as a second level of support in case the bases are not able to meet the customers' demand. The depot also serves as the distribution center for the replenishment of stock to the bases, and as the site for the repair of failed items. Due to the extensive spread of the region typically covered, a base may be closer to some neighboring bases than to the depot. In these cases, it may be more efficient for this base to seek the second level of product support from the neighboring bases, rather than the depot, when it is out of stock. Such transshipments between bases are commonly used in practice. We shall call the shipment of stock from one base to another that faces a demand when it is out of stock as *emergency lateral transshipments.*

The use of emergency lateral transshipments has the effect of reducing the time that a demand would be backordered. Consequently, the expected level and the expected cost of total backorders in the system should be reduced. Such a benefit, however, is not without a cost. First, such a practice will incur additional transportation cost for the transshipment. Other costs would also have to be included, such as the cost of issuing the lateral resupply order, and other search costs for locating among the set of neighboring bases, the one (or ones) that has (have) stock on hand. Also, by releasing an item for lateral transshipment, the neighboring base's ability to meet its customer demand is reduced.

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Significant advances have been made in modelling multi-echelon inventory systems for repairable items (see Nahmias 1981 for review). Because of the low demand and high value nature of most repairable items, a one-for-one (S-1, S) ordering policy is often used. Sherbrooke (1968) developed the well-known METRIC as an approximate model for repairable items. METRIC was later extended to the multi-indenture case by Muckstadt (1973). Simon (1971) and Shanker (1981) have given exact analyses under the assumption that repair times are deterministic. More recently, Graves (1985), Sherbrooke (1986), and Slay (1984) have proposed and tested new approximations. In none of these models is emergency lateral transshipment allowed.

Earlier inventory models that have considered transshipments between stocking locations include Gross (1963) and Das (1975). Gross's model is basically a single-echelon one with multiple stocking points. The computation involved is not very tractable with more than two locations. Das' periodic review inventory model consists of only two locations, and transshipment is only allowed at pre-specified times during the review period. Hoadley and Heyman (1977) considered a one-period multi-echelon model that allows lateral transshipment between stocking points at the same echelon level. A similar problem has been studied by Karmarkar and Patel (1977), where the problem is decomposed into a transportation problem and decoupled newsboy problems. More recently, Cohen, Kleindorfer and Lee (1986) developed a comprehensive model for a multi-echelon inventory system characterized by the existence of pooling mechanisms among locations at the same echelon level. In this work distinct groups of stocking locations are formed within which stocks can be shared. The premise is that it is more efficient to obtain needed stock from members of the so-called "pooling group" than from locations at higher echelons. In these studies, the inventory policy is periodic review, rather than the continuous one used by METRIC and other repairable inventory models. A major limitation of these models, however, is that the leadtimes for transshipment and replenishment of stock are assumed to be zero. Once this assumption is relaxed, the models become extremely complex.

Recently, Dada (1985) developed a model for a two-echelon inventory system where both the on-hand and on-order stocks at the stocking locations of the lower echelon can be pooled. Exact solutions are obtained for the case where each location stocks only one unit of the item. An approximate model is developed to compute the performance characteristics in the general case. The model assumes that *all* stocks at the lower echelon can be pooled: a restrictive assumption especially in light of the extensive geographical spread of multi-echelon systems. Moreover, all shipment times are assumed to be exponential. The model is also limited in terms of the size of the problem that can be solved. Bowman (1986) has also considered a two-echelon system with emergency lateral transshipment that is similar to that of this paper. He approximates the performance measures of the system by assuming that there is a stationary "switchboard" that reroutes demands to the lower echelon stocking points according to some equilibrium probabilities of stockouts at the stocking points. Slay (1986) independently developed a model that is very similar to the model developed here. A slightly different approximation method is used for the estimation of some key performance measures.

In this paper, we consider a multi-echelon inventory system for repairable items that employs continuous-review monitoring of inventory similar to the METRIC model of Sherbrooke (1968) and Graves (1985). A one-for-one replenishment policy is used, and emergency lateral transshipments between *some* identical bases are allowed. We focus on cases where a fairly high service level (fill-rate) is imposed at the bases, say, above 0.70. As can be expected, the exact distributions of backorders and the amount of transshipments are very complex to analyze. Nevertheless, it will be shown that simple and effective approximations for these performance measures can be readily derived. Based on the framework developed, the problem of determining optimal stocking levels

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in such a multi-echelon inventory system is also analyzed. A procedure to obtain these stocking levels is then described along with numerical examples.

The paper is organized as follows: §2 describes the model in more detail; in §3, the approximate expected backorder levels and number of emergency lateral transshipments are derived. A search procedure (with stopping rules) is given in §4 to determine the optimal stocking levels. §5 presents numerical examples illustrating the procedure and the cost savings resulted from the use of emergency lateral transshipment. §6 outlines extensions and suggestions for future research.

2. Model Description

Consider a continuous-review system with one depot and M bases as shown in Figure 1. The bases are grouped into n disjoint groups such that members of each group are identical and that they "share" their stocks. These groupings correspond to the "pooling groups" in Cohen, Kleindorfer and Lee (1986). Let m_i be the number of identical bases in pooling group i. When a member of a group is out of stock, it obtains the required unit by emergency lateral transshipment. Let T_i and a_i be the mean time and unit cost, respectively, for emergency lateral transshipment of an item among group i. These mean times and costs may differ from group to group, depending on the size and other geographical factors of the group.

When a failed item is brought to a base, if the base has stock on hand, it will replace the failed item. Simultaneously, an order is issued to the depot to replenish the base's stock. However, if the base is out of stock, it will issue an emergency transshipment request to one of the other members of its pooling group that has stock on-hand. If there is more than one base that has stock on-hand, then there are several prioritized rules to choose the one to source the emergency transshipment. The base that sources the transshipment will also issue a replenishment order to the depot to restore its inventory level. If all members of the group are out of stock, then the request is directed to the depot. In all the cases, the failed item is also sent to the depot for repair immediately.

In this paper, three prioritized source rules are considered to determine the base as the source for emergency lateral transshipment:

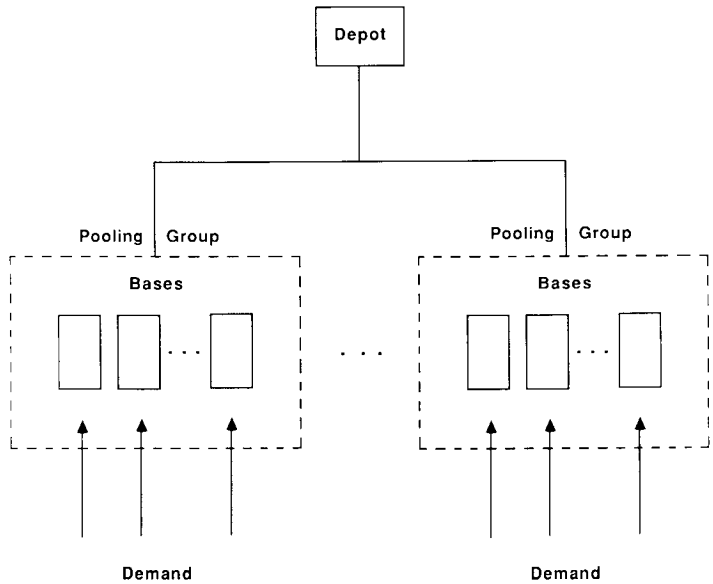

FIGURE 1. Multi-Echelon Inventory System with Emergency Lateral Transshipment.

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(i) Random source rule. The source base is chosen randomly from among the members of the pooling group with stock on hand.

(ii) Priority source rule 1. The base with the maximum stock on hand is chosen as the source base. If there are ties, then the source base is chosen randomly among the ties.

(iii) Priority source rule 2. The base with the maximum stock on hand is chosen as the source base. If there are ties, then the base with the smallest number of outstanding orders waiting at the depot is chosen. If there are still ties, then the random rule is used to break the tie.

The random source rule requires the least system status information. Priority source rule 2 would be expected to perform best, since it utilizes more system information. Obviously, there is a cost of more information. Of course, a superior policy could involve checking the inventory status in the pipeline to the base to determine the due-date of the next incoming order, and comparing it with the emergency lateral transshipment leadtime to determine whether emergency transshipment should be initiated. Such a rule requires additional system status information. By ignoring such a sourcing rule we assume that in using our rules emergency lateral transshipment would always be chosen. In repairable item systems where (S-1, S) ordering policies are used, item demand is usually very low. Hence, when the emergency transshipment times are significantly lower than the regular resupply time, it is very likely that emergency lateral transshipment is the most effective way to reduce the backorder level at the bases. We note the similarity of our assumption to that of Muckstadt and Thomas (1980), where the emergency shipment times are substantially lower than the normal resupply times.

When the lateral transshipment times are not significantly lower than the normal resupply times, emergency lateral transshipment may not be the most efficient way to operate the inventory system. If inventory status at the whole system can be obtained in real time, a dynamic rule for the dispatch of depot stock as described in Miller (1968) might be a better way to operate the system.

In our model, a demand at the base is backordered while an emergency lateral transshipment is on its way to the base (or if all members of the pooling group are out of stock and the base is waiting for the supply from the depot). Upon receiving an order from the bases, the item will then be sent out immediately, provided the depot has an inventory of repaired units on hand; otherwise, a delay will be incurred, until a repair unit becomes available at the depot.

We shall state the following additional assumptions:

(i) Bases in the same pooling group have identical repair arrival rates.

(ii) Demand at each base of pooling group i is Poisson with rate λ_i. Hence $\lambda = \sum_{i=1}^{n} m_i\lambda_i$ represents the total arrival rate to all the bases, i.e., to the depot.

(iii) The transportation time T from the depot to each site is fixed and the same for all bases.

(iv) No condemnation is assumed; i.e., all failed items can be repaired.

(v) Orders are filled on a first come, first served basis at both the depot and the bases.

(vi) Once a demand is to be met by emergency lateral transshipment, it will wait at the base until the *transshipped* unit arrives.

(vii) Repair is done at the depot, which is assumed to have an infinite number of servers; i.e., queueing does not occur at the depot.

In §6, we shall discuss how the model can be applied when assumption (iii) is relaxed. The ample-server assumption of (vii) has been used in most of the previous studies (see Sherbrooke 1968 and Graves 1985). Of particular note is assumption (vi). In actual practice, this is not strictly accurate. When a base is out of stock, there must be some units that are in resupply from the depot (unless the stocking level at this base is zero). Hence, there is a probability that the resupply unit will arrive earlier than the trans-

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shipped one, so that the demand waiting could have been met before the arrival of the transshipment.

Define:

S_i = stocking level at a base of pooling group i; $i = 1, \ldots, n$.
S_0 = stocking level at the depot.
$D(t_1, t_2)$ = total demand from all bases in $(t_1, t_2]$.
$W_i(t)$ = number of orders outstanding at a particular base of group i at time t, $i = 1, \ldots, n$.
$W'_i(t)$ = number of orders outstanding at pooling group i at time t, $i = 1, \ldots, n$.
$W_0(t)$ = number of orders outstanding at all the bases at time t.
$B_0(t)$ = number of backordered units at the depot at time t.
$R(t)$ = number of units being repaired at the depot at time t.
h = unit holding cost of the item per unit of time.
v = unit cost of a backordered item per unit of time.
$1/\mu$ = mean repair time for a failed item (may include the time to transport the failed item back to the depot).
N_i = average number of emergency lateral transshipments per unit time at pooling group i; $i = 1, \ldots, n$.

In the above definitions, an *outstanding order* is one that is due to either a delay at the repair center of the depot, or resupply in transit from the depot.

3. Operating Characteristics

The number of backorders and the quantity of emergency lateral transshipments are the two key performance measures of interest. Although the exact distributions of these two measures are difficult to obtain, approximations to their expected values can be derived.

We first observe that, regardless of whether there are emergency lateral transshipments among the bases, the arrival process of failed items and order requests to the depot is always Poisson with mean λ. Hence, the distribution of backorders at the depot is also the same. We use the result of Graves (1985) to obtain

$$W_0(t + T) = B_0(t) + D(t, t + T), \qquad \text{where} \tag{1}$$

$$B_0(t) = [R(t) - S_0]^+, \tag{2}$$

and $[x]^+$ denotes $\max[0, x]$.

By Palm's Theorem (Palm 1938), in the steady state $R(t)$ is distributed as Poisson with mean λ/μ. Furthermore, $D(t, t + T)$ is also Poisson with mean λT. Since $B_0(t)$ and $D(t, t + T)$ are independent, the steady state distribution of W_0 can be readily obtained.

We next derive approximations for the proportions: (1) β_i, that arriving demands at a base will be met immediately, and (2) α_i, that arriving demands at a base are met by emergency lateral transshipment.

To find β_i, it is useful to determine the steady state distribution of the number of orders outstanding at a base; i.e., either in emergency lateral transshipment in-transit, or depot resupply in-transit. We use the result of Simon (1971) and Graves (1985), which deal with the case with no lateral transshipments, to approximate such a distribution. In what follows, we give the rationale and motivation for such an approximation. Numerical results are also provided.

Since the arrival process of orders to the depot is Poisson (with or without lateral transshipment), we can obtain the distribution of the number of orders outstanding at pooling group i, W'_i, by treating the group as one stocking location with stock m_iS_i. In this case, the steady state probability distribution of W'_i is given by (see Graves 1985):

$$\Pr[W'_i = k] = \sum_{x=k}^{\infty} \binom{x}{k} (m_i\lambda_i/\lambda)^k (1 - m_i\lambda_i/\lambda)^{x-k} \Pr[W_0 = x]. \tag{3}$$

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Thus, $\theta_i = \Pr[W'_i \geq m_i S_i]$ represents the probability that there are at least $m_i S_i$ items outstanding at pooling group i. We will use this probability as an approximation for the probability that an arriving demand at group i will not be satisfied by stock within the group, and has to be backordered. The reason that this probability is not exact is that when $W'_i \geq m_i S_i$, there may be a base with W_i exceeding S_i, so that there would be another base with stock on-hand. We note that W_i can exceed S_i because, at the time when these demands arrive, the whole pooling group might be out of stock with base i backordered. Subsequently, other bases could have items resupplied from the depot so that the total number of outstanding items at the pooling group can be less than $m_i S_i$. The expected number of backorders at pooling group i that are to be met by the depot is $E(W'_i - m_i S_i)^+$.

Consider a particular base from pooling group i. When a random sourcing rule is used for emergency lateral transshipment, and the (identical) bases use stock level S_i, the probability that a given order among W'_i is from this particular base is $1/m_i$. For a given level of W'_i, we approximate the steady state probability distribution of W_i by:

$$\Pr[W_i = k] = \sum_{x=k}^{\infty} \binom{x}{k} (1/m_i)^k (1 - 1/m_i)^{x-k} \Pr[W'_i = x]. \tag{4}$$

Equation (4) is only an approximation since the probabilities that orders among W'_i are from a particular base are not independent. As will be seen, the derivations of α_i and β_i based on (4) are quite accurate for the other (nonrandom) sourcing rules for emergency lateral transshipment. As already noted, (4) is the same as the case where no emergency transshipments are allowed. This is because while a base may have demands referred to another base via emergency lateral transshipment, it is equally likely that demands will be referred to this base from other bases (by the same emergency lateral transshipment arrangement). Of course, (4) will not be a good approximation if emergency lateral transshipments follow a certain prespecified rule where one base is favored over another in the long run.

Based on (3) and (4), the probability β_i that an arriving demand will be met immediately by stock on-hand can be written as:

$$\beta_i = \sum_{k=0}^{S_i - 1} \sum_{x=k}^{\infty} \binom{x}{k} (\lambda_i/\lambda)^k (1 - \lambda_i/\lambda)^{x-k} \Pr[W_0 = x]. \tag{5}$$

The average number of demands per unit time met immediately at bases in group i is thus $m_i(\beta_i \lambda_i)$. The average number of demands per unit time at group i that are backordered due to lack of stock in the pooling group is $\theta_i(m_i \lambda_i)$. Hence, the average number of demands per unit time met by emergency lateral transshipments at pooling group i is $m_i\lambda_i - m_i(\beta_i\lambda_i) - \theta_i(m_i\lambda_i) = m_i\lambda_i(1 - \theta_i - \beta_i)$. The proportion of arriving demands that are met by emergency lateral transshipments is thus $\alpha_i = m_i\lambda_i(1 - \theta_i - \beta_i)/m_i\lambda_i = 1 - \theta_i - \beta_i$. The expected number of emergency lateral transshipments per unit time at pooling group i, N_i, is:

$$N_i = m_i \lambda_i \alpha_i. \tag{6}$$

The computation of β_i and α_i requires derivation of (3) and (5), via (1) and (2). In practice, this could be a cumbersome task. We use the negative binomial approximations as suggested by Graves (1985), Sherbrooke (1986), Slay (1984), and Lee and Moinzadeh (1987). We conducted extensive tests for the goodness of the approximations for β_i and α_i using computer simulation. Since we are interested in systems that give fairly good service levels, we tested only those cases where β_i is above 70%. The

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simulation was conducted with λ_i identical for all bases, $T = 2$, and the repair time a random variable with a fixed component 2 (representing the fixed shipment time from the base back to the depot) and a variable component exponentially distributed with mean 2. The pooling group under consideration had 3 bases, and $\lambda = 13\lambda_i$. We note that, to obtain the operating characteristics of a particular pooling group, it is immaterial to know how the other bases are grouped. All three sourcing rules for the selection of the source base for emergency lateral transshipment were separately simulated. Results are compared to the approximate model results. Tables 1 and 2 show the results of the different values of λ_i, S_i and S_0 tested.

In most cases the approximation is accurate to 2 decimal places. The differences between the three sourcing rules are not significant, probably due to the low-demand nature of the items. Experimentation with $m_i = 2$ and $m_i = 4$ showed similar results. It should be noted, however, that for low service-levels ($\beta_i \ll 0.7$), the performance of the approximations deteriorates. For $\beta_i \geq 0.7$, we concluded that the approximations are very good.

Little's formula (Little 1961) provides the expected number of items in transit for emergency lateral transshipments as $N_i T_i$.

Finally, the total number of backorders at pooling group i, $B_i(t)$, consists of the number of backorders that are to be met by the depot, and the number of backorders that are to be met by emergency lateral transshipments, i.e., items in-transit of emergency lateral transshipments. Hence,

$$E(B_i) = E(W'_i - m_i S_i)^+ + N_i T_i. \tag{7}$$

4. Determination of Optimal Stocking Levels

A general objective function for the management of the multi-echelon inventory system is to minimize total cost, composed of inventory holding cost, backorder cost, and the cost of emergency lateral transshipments subject to some service level constraints. This can be stated as (see Richards 1976 and Lee and Moinzadeh 1987):

$$\text{Minimize } C = h(S_0 + \sum_{i=1}^{n} m_i S_i) + v \sum_{i=1}^{n} E(B_i) + \sum_{i=1}^{n} a_i N_i, \tag{8}$$

subject to:

Immediate fill-rate at a base of pooling group $i \geq \xi_i$;

Fill-rate after emergency lateral transshipment at bases of pooling group $i \geq \xi'_i$;

Fill-rate at the depot $\geq \xi_0$.

The service level constraints are equivalent to setting:

$$\beta_i \geq \xi_i, \qquad i = 1, \ldots, n; \tag{9}$$

$$\beta_i + \alpha_i = 1 - \theta_i \geq \xi'_i, \qquad i = 1, \ldots, n; \qquad \text{and} \tag{10}$$

$$\Pr[R \geq S_0] \geq \xi_0. \tag{11}$$

Constraint (11) gives a lower bound on the level of S_0 at the depot. For a given S_0, (9) and (10) also jointly define a lower bound on the level of S_i. Let $\underline{S}_i$ be the smallest S_i that satisfies (9) and (10).

Our approach to the problem of determining the optimal stocking levels is a two-phase method similar to that of Sherbrooke (1968) and Muckstadt (1978). For each given S_0, we find the S_i's which minimize C. The second phase optimizes S_0.

For a given S_0, the optimal S_i is the one that minimizes

$$\begin{aligned} C_i(S_i|S_0) &= hm_i S_i + vE(B_i) + a_i N_i \\ &= hm_i S_i + vE(W'_i - m_i S_i)^+ + (vT_i + a_i)m_i \lambda_i \alpha_i. \end{aligned} \tag{12}$$

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TABLE 1

Evaluating the Effectiveness of β_i

λ_i	S_0	S_i	Approximation	Simulation: Random Sourcing Rule	Simulation: Priority Rule (1)	Simulation: Priority Rule (2)
0.04	1	1	0.84	0.84	0.84	0.84
		2	0.99	0.98	0.99	0.99
	2	1	0.88	0.90	0.90	0.90
		2	0.99	0.99	1.00	1.00
0.06	2	1	0.80	0.80	0.79	0.79
		2	0.98	0.98	0.98	0.98
	4	1	0.86	0.87	0.87	0.87
		2	0.99	0.99	0.99	0.99
0.08	4	1	0.80*†#	0.78	0.78	0.78
		2	0.98	0.98	0.98	0.98
	6	1	0.84	0.84	0.84	0.84
		2	0.99	0.99	0.99	0.99
0.1	4	1	0.73*†	0.70	0.69	0.69
		2	0.96	0.96	0.96	0.96
		3	0.99	0.99	1.00	1.00
	6	1	0.79	0.77	0.76	0.76
		2	0.97	0.98	0.98	0.98
		3	1.00	1.00	1.00	1.00
0.2	5	2	0.80*	0.79	0.79	0.79
		3	0.95	0.95	0.95	0.95
		4	0.99	0.99	0.99	0.99
	10	2	0.90	0.90	0.90	0.90
		3	0.98	0.98	0.99	0.99
		4	1.00	1.00	1.00	1.00
0.4	12	3	0.81*	0.79	0.81	0.82
		4	0.93†#	0.93	0.95	0.95
	18	3	0.90	0.90	0.91	0.91
		4	0.97	0.97	0.98	0.98
	30	2	0.81*	0.79	0.80	0.80
		3	0.95	0.95	0.96	0.96
		4	0.99	0.99	0.99	0.99
0.6	14	4	0.75*†	0.73	0.73	0.74
		5	0.88†#	0.89	0.90	0.90
		6	0.95	0.96	0.96	0.96
		7	0.98	0.99	0.99	0.99
	21	4	0.85	0.86	0.86	0.86
		5	0.94	0.95	0.95	0.95
		6	0.98	0.98	0.98	0.98
	35	3	0.86	0.86	0.87	0.87
		4	0.96	0.96	0.96	0.96
		5	0.99	0.99	0.99	0.99

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TABLE 1 (*cont'd*)

λ_i	S_0	S_i	Approximation	Simulation: Random Sourcing Rule	Simulation: Priority Rule (1)	Simulation: Priority Rule (2)
0.8	20	5	0.77	0.76	0.76	0.77
		6	0.88	0.89	0.90	0.90
		7	0.95	0.95	0.95	0.96
		8	0.98	0.98	0.98	0.98
	30	4	0.75*†#	0.73	0.74	0.74
		5	0.88	0.89	0.90	0.90
		6	0.95	0.95	0.96	0.96
		7	0.98	0.99	0.99	0.99
	40	4	0.87	0.88	0.88	0.88
		5	0.95	0.96	0.96	0.96
		6	0.99	0.98	0.98	0.99

Key: * outside 95% confidence interval of simulation result under Random Dispatch Rule.
† outside 95% confidence interval of simulation result under Priority Rule (1).
# outside 95% confidence interval of simulation result under Priority Rule (2).

Define

$$\phi_i(S_i) = E[W'_i - m_i(S_i + 1)]^+ - E[W'_i - m_i S_i]^+$$
$$= -m_i \Pr[W'_i \geq m_i(S_i + 1)] - \sum_{k=m_i S_i}^{m_i(S_i+1)-1} (k - m_i S_i) \Pr[W'_i = k]. \tag{13}$$

The following proposition gives an upper bound for the optimal value of S_i for a given S_0.

PROPOSITION 1. *For a given S_0, suppose $\underline{S}_i$ is such that* $\Pr[W_i = k]$ *as given by* (4) *and* $\Pr[W'_i = k]$ *as given by* (3) *are nonincreasing in k, for $k \geq \underline{S}_i$ and $k \geq m_i \underline{S}_i$, respectively. If $\bar{S}_i$ is such that:*

$$h - v\phi_i(\bar{S}_i) - (vT_i + a_i)m_i\lambda_i \Pr[W_i = \bar{S}_i] \geq 0, \tag{14}$$

then all S_i greater than $\bar{S}_i$ cannot be optimal.

In general, using the negative binomial approximation, the probability functions of W_i and W'_i are unimodal and decreasing in their tails. As Muckstadt (1978) has observed, in most practical situations, the penalty on backorders and the service-level constaints result in the tails of the distributions as the only important portion of the distributions. Hence, it is reasonable to expect the probability functions of W_i and W'_i to be nonincreasing in the relevant range considered.

Define the optimal cost of C_i for a given S_0, as: $C_i^*(S_0) = \min_{S_i} C_i(S_i|S_0)$.

For a given S_0, we can search over S_i to obtain the optimal cost. The second phase of the search is to look for the optimal value of S_0. To find a stopping rule for the search of the optimal value of S_0, we first note that, for $S'_0 < S''_0$,

$$C_i^*(S'_0) \geq C_i^*(S''_0). \tag{15}$$

Recall from (1) that, since $S'_0 < S''_0$, $W_0|S'_0$ stochastically dominates $W_0|S''_0$. As proven by Moinzadeh and Lee (1986), this implies that $W'_i|S'_0$ and $W_i|S'_0$ also stochastically dominate $W'_i|S''_0$ and $W_i|S''_0$, respectively. As expected, when we have more stock at the

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TABLE 2

Evaluating the Effectiveness of α_i

λ_i	S_0	S_i	Approximation	Simulation: Random Sourcing Rule	Simulation: Priority Rule (1)	Simulation: Priority Rule (2)
0.04	1	1	0.14	0.14	0.15	0.15
		2	0.01	0.02	0.01	0.02
	2	1	0.11	0.10	0.10	0.10
		2	0.01	0.01	0.00	0.00
0.06	2	1	0.16	0.18	0.18	0.18
		2	0.02	0.02	0.02	0.02
	4	1	0.12	0.12	0.12	0.12
		2	0.01	0.01	0.01	0.01
0.08	4	1	0.16*†	0.19	0.19	0.19
		2	0.02	0.02	0.02	0.02
	6	1	0.14	0.15	0.15	0.15
		2	0.01	0.01	0.01	0.01
0.1	4	1	0.19*†	0.24	0.24	0.24
		2	0.04	0.04	0.04	0.04
		3	0.00	0.00	0.00	0.00
	6	1	0.17	0.19	0.20	0.20
		2	0.03	0.02	0.02	0.02
		3	0.00	0.00	0.00	0.00
0.2	5	2	0.15*	0.17	0.18	0.18
		3	0.05	0.05	0.04	0.04
		4	0.01	0.01	0.01	0.01
	10	2	0.09	0.09	0.09	0.09
		3	0.02†	0.02	0.01	0.01
		4	0.00	0.00	0.00	0.00
0.4	12	3	0.14*	0.16	0.16	0.15
		4	0.06†#	0.06	0.05	0.05
	18	3	0.09	0.09	0.08	0.09
		4	0.03	0.03	0.02	0.02
	30	2	0.16*†#	0.18	0.18	0.18
		3	0.05	0.05	0.04	0.04
		4	0.01	0.01	0.01	0.01
0.6	14	4	0.16*†#	0.19	0.19	0.18
		5	0.10	0.10	0.09	0.09
		6	0.05	0.04	0.04	0.04
		7	0.02	0.01	0.01	0.01
	21	4	0.12	0.12	0.12	0.12
		5	0.05	0.05	0.04	0.05
		6	0.02	0.02	0.02	0.02
	35	3	0.12	0.12	0.12	0.12
		4	0.04	0.04	0.04	0.04
		5	0.01	0.01	0.01	0.01

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TABLE 2 (*cont'd*)

λ_i	S_0	S_i	Approximation	Simulation: Random Sourcing Rule	Simulation: Priority Rule (1)	Simulation: Priority Rule (2)
0.8	20	5	0.15*†	0.17	0.17	0.17
		6	0.10	0.10	0.09	0.09
		7	0.05	0.05	0.05	0.04
		8	0.02	0.02	0.02	0.02
	30	4	0.15*†	0.19	0.19	0.18
		5	0.10	0.10	0.09	0.09
		6	0.04	0.04	0.04	0.04
		7	0.02	0.01	0.01	0.01
	40	4	0.11	0.11	0.10	0.10
		5	0.04	0.04	0.04	0.04
		6	0.01	0.02	0.02	0.01

Key: * outside 95% confidence interval of simulation result under Random Dispatch Rule.
† outside 95% confidence interval of simulation result under Priority Rule (1).
# outside 95% confidence interval of simulation result under Priority Rule (2).

depot, there would be fewer—in a stochastic dominance sense—items in resupply or outstanding at the bases. Hence (15) holds.

As a result, an obvious lower bound for $C_i(S_i|S_0')$ is $C_i^*(S_0 \to \infty)$. When $S_0 \to \infty$, B_0 from (1) is always zero, and so W_0 is simply Poisson with mean λT. Consequently, W_i' and W_i are Poisson with mean $m_i\lambda_i T$ and $\lambda_i T$, respectively. Using the procedure described above, we can find the S_i's that minimize $C_i(S_i|S_0 \to \infty)$, for all $i = 1, \ldots, n$. Let $\underline{C} = \Sigma_{i=1}^n C_i^*(S_0 \to \infty)$.

The following proposition gives a stopping rule for the search of the optimal value of S_0. The proofs of both Propositions 1 and 2 are given in the Appendix.

PROPOSITION 2. *Suppose $\bar{S}_0$ is such that*

$$h \geq \sum_{i=1}^{n} C_i^*(\bar{S}_0) - \underline{C}, \tag{16}$$

then all $S_0 > \bar{S}_0$ cannot be optimal.

5. Numerical Examples

To illustrate the application of the above procedure for the determination of optimal stocking levels, the following numerical example is presented:

$n = 3$;
$m_1 = 3$, $m_2 = 2$, $m_3 = 3$;
$\lambda_1 = 0.1$, $\lambda_2 = 0.1$, $\lambda_3 = 0.25$;
$\xi_i = 0.7$, for all i;
$\xi_i' = 0.85$, for all i;
$\xi_0 = 0$;
$T = 3$;
$T_i = 1$, for all i;
$v = 60$;
$a_i = 30$, for all i;
$h = 20$;
$1/\mu = 4$.

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The cost function used here is based on the approximate model. As described in the last section, we first obtain $\underline{C}$ as 240.89 (note that $\underline{C}$ does not include the holding cost of stock at the depot).

The stopping-rule of Proposition 2 shows that it is sufficient to consider S_0 up to $S_0 = 6$. This is because at $S_0 = 6$,

$$\sum_{i=1}^{n} C_i^*(S_0 = 6) - \underline{C} = 244.41 - 240.89 \leq h.$$

It can thus be seen that the stopping-rule could be very effective in limiting the search of the optimal stocking levels in the system.

The optimal stocking levels for this example are: $S_0 = 6$, $S_1 = 1$, $S_2 = 1$, $S_3 = 2$. The corresponding optimal total cost here, where emergency lateral transshipment is used, is 364.41.

When no lateral transshipment is allowed, the specification of the base fill-rates that correspond to $\xi = 0.7$ and $\xi' = 0.85$ in the lateral transshipment case is not trivial. The appropriate specification of fill-rates should probably be between 0.7 and 0.85. The optimal stocking levels for different specifications of the fill-rates are obtained. For example, if the fill-rates are specified to be 0.75, then the optimal stocking levels are given by: $S_0 = 1$, $S_1 = 2$, $S_2 = 2$, $S_3 = 3$, and the corresponding optimal total cost is 427.99. Using these stocking levels but allowing lateral transshipment would result in a total cost of 418.59.

As can be seen, the use of emergency lateral transshipment can result in significant savings. In the above example, the optimal base-stock levels without transshipment are higher than that with transshipment, whereas the reverse is true for the depot stock level. With emergency lateral transshipments, less stock is needed at the bases, since sharing is possible at the base level. In general, the problem of whether emergency lateral transshipment should be used or not depends on the magnitudes of the costs and the leadtimes of transshipments.

Additional examples with $h = 10$ and $T_i = 0.25$ or 1 have also been run, and their results are summarized in Table 3 and Table 4. Of interest is the performance of the stocking levels derived from the use of traditional METRIC models, i.e., without emergency lateral transshipment. The resulting optimal stocking levels from METRIC depend on how one specifies the fill-rate that corresponds to the combination of ξ_i and ξ'_i, i.e., the base and pooling group fill-rates respectively. In the particular examples examined, it turns out that by setting the fill-rate to correspond to $\xi_i = 0.7$ in METRIC no cost penalty is incurred. In general, however, such specification may not yield stocking levels that would provide a service that is equivalent to $\xi_i = 0.7$ and $\xi'_i = 0.85$. Setting the fill-rate higher than 0.7 would result in cost penalties as high as 25%. Of course, the cost penalty in general depends highly on the data parameters used. The results here are for illustrative purposes only.

6. Conclusion

Emergency lateral transshipments have often been used in multi-location inventory systems to improve service performance. In this paper, we have presented a multi-echelon repairable inventory system where emergency lateral transshipments are allowed. Approximations for the expected values of performance measures such as the back-order level and the number of emergency lateral transshipments are derived. A procedure is also given for the determination of the optimal stocking levels in such a system.

In this paper, we have assumed that the shipment times from the depot to the bases are the same for all bases. Such an assumption can actually be relaxed, as in Graves (1985). Suppose T'_i is the shipment time from the depot to pooling group i, $i = 1, \ldots, n$.

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TABLE 3

Examples of Optimal Stocking Levels and Costs Without Emergency Lateral Transshipment

	h	10		20	
	T_i	0.25	1	0.25	1
(i) $\xi = 0.85$	Optimal Stocking Levels#	(3; 2, 2, 3)		(3; 2, 2, 3)	
	Expected Cost	233.63		453.63	
(ii) $\xi = 0.8$	Optimal Stocking Levels	(2; 2, 2, 3)		(2; 2, 2, 3)	
	Expected Cost	229.80		439.80	
(iii) $\xi = 0.75$	Optimal Stocking Levels	(1; 2, 2, 3)		(1; 2, 2, 3)	
	Expected Cost	227.99		427.99	
(iv) $\xi = 0.72$	Optimal Stocking Levels	(7; 1, 1, 2)		(7; 1, 1, 2)	
	Expected Cost	205.09		385.09	
(v) $\xi = 0.7$	Optimal Stocking Levels	(6; 1, 1, 2)		(6; 1, 1, 2)	
	Expected Cost	199.06		369.06	

The optimal stocking levels are stated as $(S_0; S_1, S_2, S_3)$.

TABLE 4

Examples of Optimal Stocking Levels and Costs with Emergency Lateral Transshipment

	h	10		20	
	T_i	0.25	1	0.25	1
(0) With Emergency Lateral Transshipment $\xi = 0.7$, $\xi' = 0.85$	Optimal Stocking Levels#	(6; 1, 1, 2)		(6; 1, 1, 2)	
	Expected Cost	185.22	194.41	355.22	364.41
Expected Cost Using Stocking Levels of Table 3 with Emergency Lateral Transshipment	(i)	226.32	231.47	4[illegible]32	451.47
	(ii)	218.43	224.73	428.43	434.73
	(iii)	211.10	218.59	411.10	418.59
	(iv)	193.61	202.66	373.61	382.66
	(v)	185.22	194.41	355.22	364.41
Cost Penalty (%) of Using Stocking Levels of Table 3 with Emergency Lateral Transshipment over that of (0) above	(i)	22.2	19.1	25.6	23.9
	(ii)	17.9	15.6	20.6	19.3
	(iii)	14.0	12.4	15.7	14.9
	(iv)	4.5	4.2	5.2	5.0
	(v)	0.0	0.0	0.0	0.0

The optimal stocking levels are stated as $(S_0; S_1, S_2, S_3)$.

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Then we can obtain $W'_i(t)$ from: $W'_i(t + T'_i) = B_i(t) + D_i(t, t + T'_i)$, where $B_i(t)$ is the number of backordered units at the depot at time t that originated from pooling group i, and $D_i(t, t + T'_i)$ is the demand of pooling group i in $(t, t + T'_i]$, which is Poisson with mean $m_i\lambda_i T'_i$. The steady state distribution of B_i can be found by a similar disaggregation scheme as in (3), once that of B_0 is known.

Incorporation of emergency lateral transshipment is an important problem in multi-echelon inventory models. This paper has attempted to extend the basic METRIC framework to cases when emergency lateral transshipments are allowed and used. In general, the factors that determine the grouping of bases could include distance, accessibility, extent to which inventory status information can be shared, and other managerial considerations. Design issues such as how emergency lateral transshipment should be used when the transshipment leadtimes are significant; how bases should be grouped to form the pooling groups; operational issues like other sourcing rules for emergency lateral transshipment; and the extension of the current model to consumable items with an external supplier, are possible topics for future research.[1]

[1] The author would like to thank George Tagaras for writing and executing the computer simulation in §3. Helpful comments by Professor Warren Hausman, the anonymous referees and the Departmental Editor are also gratefully acknowledged.

Appendix

PROOF OF PROPOSITION 1. The first term of $\phi_i(S_i)$ in (13) is clearly nondecreasing in S_i. Next, for any $S_i \geq \underline{S}_i$,

$$-\sum_{k=m_i(S_i+1)}^{m_i(S_i+2)-1} [k - m_i(S_i + 1)] \Pr[W'_i = k] + \sum_{k=m_iS_i}^{m_i(S_i+1)-1} (k - m_iS_i) \Pr[W'_i = k]$$

$$= \sum_{k=m_iS_i}^{m_i(S_i+1)-1} (k - m_iS_i)\{\Pr[W'_i = k] - \Pr[W'_i = k + m_i]\} \geq 0,$$

by the supposition that $\Pr[W_i = k]$ is nonincreasing in k for $k \geq m_i\underline{S}_i$.

Hence, $\phi_i(S_i)$ is nondecreasing in S_i. For any $S_i \geq \bar{S}_i$, we thus have

$$C_i(S_i + 1|S_0) - C_i(S_i|S_0) = h - v\phi_i(S_i) + (vT_i + a_i)m_i\lambda_i\{-\Pr[W_i = S_i] + \sum_{k=0}^{m_i-1} \Pr[W'_i = m_iS_i + k]\}$$

$$\geq h - v\phi_i(\bar{S}_i) - (vT_i + a_i)m_i\lambda_i \Pr[W_i = \bar{S}_i] \geq 0.$$

The second last inequality above is due to the supposition that $\Pr[W_i = k]$ is nonincreasing in k, for $k \geq \underline{S}_i$. Q.E.D.

PROOF OF PROPOSITION 2. By (15) and the definition of $\underline{C}$, the maximum reduction in the cost function $\Sigma_{i=1}^n C_i(S_i|S_0)$ by incrementing $\bar{S}_0$ by one is at most the L.H.S. of (16). The additional holding cost for increasing depot stock by one is h. Hence, if (16) holds, it is not worthwhile to increase $\bar{S}_0$ by one. Similarly, it is not worthwhile to increase $\bar{S}_0$ by two, . . . , and so on. Q.E.D.

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